

Bachelor of Technology — Data Science and Artificial Intelligence

Table 1: Data Science and Artificial Intelligence — B25 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	33
	Discipline Electives	33
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 2: B. Tech Data Science and Artificial Intelligence — B25 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CS-213	Reverse Engineering	0-0-2-1	1
2	DS-314*	Probability Theory and Statistics	3-1-0-4	4
3	DS-412	Matrix Computations for Data Science	3-0-2-4	4
4	DS-312*	Computing Systems for Data Processing	3-0-2-4	4
5	DS-415*	Optimization for Machine Learning	3-0-2-4	4
6	DS-417 [†]	Deep Learning	3-0-2-4	4
7	CS-315*	Artificial Intelligence	3-0-2-4	4
8	DS-416*	Fundamentals of Machine Learning	3-0-2-4	4
9	DS-418 [†]	Introduction to Generative AI	3-0-2-4	4
Total				33

* Recoded w.e.f. batch 2025: CS-305 → CS-315 (Artificial Intelligence) and DS-302 → DS-312 (Computing Systems for Data Processing), both now 3-0-2-4; DS-313 → DS-314 (Probability Theory and Statistics, 3-1-0-4 unchanged); DS-411 → DS-415 (Optimization for Machine Learning) and DS-413 → DS-416 (Fundamentals of Machine Learning), both now 3-0-2-4.

[†]DS-417 (Deep Learning) and DS-418 (Introduction to Generative AI) are DSAI-only codes for the 600-level courses CS-671 and CS-683; DSAI students cannot also take CS-671 or CS-683 as electives. DS-417 is taken in Semester 4 (moved from Semester 6).

Table 3: B. Tech Data Science and Artificial Intelligence — B25 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-112	Calculus	1.5-0.5-0-2	2
3	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
4	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
5	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
6	IC Basket (choose 1)	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
7		IC-136	Understanding Biotechnology and its Applications	3-0-0-3	
8		IC-230	Environmental Science	3-0-0-3	
9	IKS + HSS	IC-182	History of Science and Technology	3	3
Total					22

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 4: B. Tech Data Science and Artificial Intelligence — B25 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-252	Probability and Statistics	3-0-2-4	4
7	IC Basket	IC-253	Data Structures and Algorithms	2-0-2-3	3
Total					20

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 5: B. Tech Data Science and Artificial Intelligence — B25 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-222P	Physics Practicum	0-0-3-2	2
3	IC	IC-272	Machine Learning	2-0-2-3	3
4	DC	CS-213	Reverse Engineering	0-0-2-1	1
5	DC	DS-314*	Probability Theory and Statistics	3-1-0-4	4
6	DC	DS-412	Matrix Computations for Data Science	3-0-2-4	4
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 6: B. Tech Data Science and Artificial Intelligence — B25 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	DS-312*	Computing Systems for Data Processing	3-0-2-4	4
2	DC	DS-415*	Optimization for Machine Learning	3-0-2-4	4
3	DC	DS-417 [†]	Deep Learning	3-0-2-4	4
Total					12

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 7: B. Tech Data Science and Artificial Intelligence — B25 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-315*	Artificial Intelligence	3-0-2-4	4
2	DC	DS-416*	Fundamentals of Machine Learning	3-0-2-4	4
Total					8

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 8: B. Tech Data Science and Artificial Intelligence — B25 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	DS-418 [†]	Introduction to Generative AI	3-0-2-4	4
2	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					8

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 9: B. Tech Data Science and Artificial Intelligence — B25 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	DS-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 10: B. Tech Data Science and Artificial Intelligence — B25 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	DS-010	Internship		2
2	MTP	DS-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 33 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 11: Data Science and Artificial Intelligence — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	AR-507	Probabilistic Robotics	3
2	AR-516	Introduction to Blockchain and Web3	3
3	AR-517	Introduction to Cyber security	3
4	BE-303	Applied Biostatistics	4
5	BE-304	Bioinformatics	4
6	BE-502	Design and Analysis of Bioalgorithms	3
7	CS-212	Design and Analysis of Algorithms	4
8	CS-214	Computer Organization	4
9	CS-241	Introduction to Cryptography	3
10	CS-302	Paradigms of Programming	4

Data Science and Artificial Intelligence — Available Discipline Electives (choose from these)
(continued)

Sl. No	Course Code	Course Name	Credits
11	CS-303	Software Engineering	3
12	CS-304	Formal Languages and Automata Theory	3
13	CS-306	Operating Systems	3
14	CS-307	Systems Practicum	2
15	CS-308	Large Application Practicum	2
16	CS-308P	Large Application Practicum	2
17	CS-309	Information and Database Systems	4
18	CS-310	Introduction to Communication and Distributed Processes	4
19	CS-313	Computer Networks	4
20	CS-347	Software Engineering	4
21	CS-350	Computer Graphics	3
22	CS-401	Architecture of High Performance Computers	3
23	CS-402	Compilers	2
24	CS-403	Algorithm Design and Analysis I	3
25	CS-405	Verification of Reactive Systems	3
26	CS-406	Computer Networks	4
27	CS-451	Computer Graphics and Game Design	3
28	CS-456	Distributed Databases	3
29	CS-501	Access Networks	3
30	CS-502	Compiler Design	4
31	CS-506	Cognitive Modeling	3
32	CS-507	Computer Architecture	4
33	CS-508	Introduction to Network Security	3
34	CS-510	Randomized Algorithms	4
35	CS-511	Applied Probability	2
36	CS-512	Matrix Theory	2
37	CS-513	Discrete Mathematics	4
38	CS-514	Advance Data Structure and Algorithm (CS-580)	4
39	CS-515	Advanced Computer Science Practicum	3
40	CS-516	Exploratory Project	3
41	CS-520	Introduction to Quantum Computing	3
42	CS-521	Introduction to Post Quantum Security	3
43	CS-522	Distributed Algorithms	3
44	CS-524	Computational Complexity Theory	4
45	CS-525	Computational Foundations of Cryptography	3
46	CS-526	Mathematical Foundation of Cryptography	3
47	CS-528	Unconditional Lower Bounds for Algorithms and Data Structures	3
48	CS-541	IoT systems and the Cloud	3

Data Science and Artificial Intelligence — Available Discipline Electives (choose from these)
(continued)

Sl. No	Course Code	Course Name	Credits
49	CS-542	Design Patterns for Scalable Systems	3
50	CS-544	Formal Concept Analysis: Theory and Practice	3
51	CS-545	Software Design Patterns	3
52	CS-546	Design of Concurrent Software	3
53	CS-547	Network Management Systems	3
54	CS-548	Cloud Networking	3
55	CS-549	Performance Analysis of Computer Networks	3
56	CS-550	Computer Graphics and Geometric Design	3
57	CS-551	Human Computer Interaction	3
58	CS-560	Text Retrieval and Mining	3
59	CS-561	Map Reduce and Big Data	3
60	CS-563	Scalable Data Science	4
61	CS-570	Algorithms for Curves and Surface	3
62	CS-571	Programming Practicum	3
63	CS-591	Special Topics in Knowledge Discovery and Data	1
64	CS-592	Selected Topics in Artificial Intelligence	2
65	CS-601	Probability and Random Processes	4
66	CS-603	Managerial Decision Making	3
67	CS-606	Cognitive Modeling	3
68	CS-609	Speech processing	3
69	CS-611	Program Analysis	4
70	CS-630	Speech Technology	4
71	CS-660	Data Mining for Decision Making	4
72	CS-661	Knowledge Representation and Reasoning	3
73	CS-662	Mobile Virtual Reality and Artificial Intelligence	3
74	CS-670	Kernal Methods for Pattern Recognition	4
75	CS-672	Advanced Topics in Deep Learning	4
76	CS-673	Advanced Computer Vision	4
77	CS-677	Soft Computing	3
78	CS-685	Natural Language Processing	3
79	CS-686	Large Language Models: Core Concepts to Custom	4
80	CS-693	Compressed Sensing and its applications	3
81	EE-305	Digital Signal Processing	4
82	EE-511	Computer Vision	4
83	EE-530	Optimization theory	3
84	EE-531	Estimation and Detection Theory	3
85	EE-541	Tensors: Techniques, Algorithms, Applications for Sig- nal	4
86	EE-543	Vision and Learning Based Control	3
87	EE-574	Biomedical Signal and Image Analysis	3

Data Science and Artificial Intelligence — Available Discipline Electives (choose from these)
(continued)

Sl. No	Course Code	Course Name	Credits
88	EE-575	Applied Statistics for Data and Signals	4
89	EE-608	Digital Image Processing	4
90	EE-620	Advanced Digital Signal Processing	3
91	ME-522	High-Performance Scientific Computing	3
<i>Compulsory project components — counted as DE, not selectable</i>			
92	DS-498P	Major Technical Project - I (compulsory; counted as DE)	4
93	DS-499P	Major Technical Project - II (compulsory; counted as DE)	4